Variation of Parameters

ANSWER KEY

Section 1: The Wronskian Determinant

1 What is the definition of the Wronskian $W(y_1, y_2)$ of two differentiable functions $y_1(x)$ and $y_2(x)$?

a. $W(y_1, y_2) = y_1 y_2' - y_1' y_2 = \det[[y_1, y_2], [y_1', y_2']]$ ✓

b. $W(y_1, y_2) = y_1 y_2 + y_1' y_2'$

c. $W(y_1, y_2) = y_1' y_2 - y_1 y_2'$

d. $W(y_1, y_2) = \frac{y_1}{y_2}$

The Wronskian is the determinant of the fundamental matrix: $W = \det[[y_1, y_2], [y_1', y_2']] = y_1 y_2' - y_1' y_2$.

2 Compute the Wronskian of $y_1 = \cos x$ and $y_2 = \sin x$.

a. 1 ✓

b. 0

c. -1

d. $\cos(2x)$

$W = (\cos x)(\cos x) - (-\sin x)(\sin x) = \cos^2 x + \sin^2 x = 1$.

3 What does $W(y_1, y_2) \neq 0$ guarantee about solutions y_1 and y_2 to a homogeneous linear ODE?

a. **They are linearly independent and form a fundamental set of solutions** ✓

b. They are orthogonal

c. They are periodic

d. One must be zero

If the Wronskian is non-zero at a point in the interval, Abel's theorem guarantees it is non-zero everywhere, establishing linear independence.

Section 2: The Variation of Parameters Formula

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4 For the second-order equation $y'' + P(x)y' + Q(x)y = g(x)$, what is the variation of parameters formula for y_p?

a. $y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx$ ✓

b. $y_p = y_1 \int \frac{y_2 g}{W} dx - y_2 \int \frac{y_1 g}{W} dx$

c. $y_p = \int \frac{g(x)}{W} dx$

d. $y_p = -y_2 \int \frac{y_1}{W} dx$

Setting $y_p = u_1 y_1 + u_2 y_2$ yields $u_1' = -\frac{y_2 g}{W}$ and $u_2' = \frac{y_1 g}{W}$, leading directly to $y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx$.

5 What must the coefficient of y'' be before applying the variation of parameters formula?

a. **Exactly `1` (must be in standard monic form)** ✓

b. Any non-zero constant

c. Zero

d. Positive

The standard derivation requires the leading coefficient to be normalized to `1`; otherwise $g(x)$ must be divided by the coefficient of y''.

6 For $y'' + y = \sec x$, where $y_1 = \cos x, y_2 = \sin x$, what is $u_1' = -\frac{y_2 g}{W}$?

a. $-\tan x$ ✓

b. $-\sin x$

c. -1

d. $-\cos x$

With $W = 1$ and $g = \sec x$: $u_1' = -\frac{\sin x \sec x}{1} = -\frac{\sin x}{\cos x} = -\tan x$.

Section 3: Solving Advanced Nonhomogeneous Equations

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7 Find $u_1 = \int(-\tan x)dx$ for the problem $y'' + y = \sec x$.

a. $\ln|\cos x|$ ✓

b. $-\ln|\cos x|$

c. $\sec^2 x$

d. $-\cos x$

$u_1 = -\int \tan x dx = -(-\ln|\cos x|) = \ln|\cos x|$.

8 Find $u_2' = \frac{y_1 g}{W}$ for $y'' + y = \sec x$ and evaluate u_2.

a. $u_2 = x$ ✓

b. $u_2 = \sin x$

c. $u_2 = \cos x$

d. $u_2 = \tan x$

$u_2' = \frac{\cos x \sec x}{1} = 1 \Rightarrow u_2 = \int 1 dx = x$.

9 Assemble the particular solution $y_p = u_1 y_1 + u_2 y_2$ for $y'' + y = \sec x$.

a. $y_p = \cos x \ln|\cos x| + x \sin x$ ✓

b. $y_p = \cos x + x \sin x$

c. $y_p = \ln|\cos x| + x$

d. $y_p = \sin x \ln|\cos x| + x \cos x$

Combining $u_1 = \ln|\cos x|$ and $u_2 = x$: $y_p = u_1 y_1 + u_2 y_2 = \cos x \ln|\cos x| + x \sin x$.

Section 4: Comparison of Nonhomogeneous Methods

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10 What is the major advantage of Variation of Parameters over Undetermined Coefficients?

a. **It works for any continuous forcing function $g(x)$, not just polynomials, exponentials, or sines/cosines ✓**

b. It never requires any integration

c. It works without knowing the homogeneous solution y_c

d. It always produces algebraic rather than transcendental answers

Variation of Parameters is completely general and applies to any continuous $g(x)$ (such as $\sec x, \tan x, 1/x$), provided y_c is known.

11 What is the main drawback of Variation of Parameters compared to Undetermined Coefficients?

a. **The integrals $\int \frac{y_i g}{W} dx$ may be difficult or impossible to evaluate in closed elementary form ✓**

b. It only works on first-order equations

c. It gives only approximate numerical solutions

d. It cannot handle initial value problems

Variation of Parameters often produces non-elementary integrals that cannot be expressed using standard functions.

12 What is Abel's formula for the Wronskian of solutions to $y'' + P(x)y' + Q(x)y = 0$?

a. $W(x) = Ce^{-\int P(x)dx}$ ✓

b. $W(x) = Ce^{\int Q(x)dx}$

c. $W(x) = C(P(x) + Q(x))$

d. $W(x) = Cx$

Abel's Identity states that the Wronskian satisfies the first-order differential equation $W' + P(x)W = 0$, giving $W(x) = Ce^{-\int P(x)dx}$.

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